THE UNFINISHED WORK

An Unfinished Symphony: Completing the Playbook for Reading Education

Why researchers, teachers, and writers now argue America's reading crisis is a **professionalization problem — not a knowledge one.**

For many, the Science of Reading movement has entered its 'trough of despair.' Not only are nationwide scores not up; they are, in fact, down. Finger-pointing has begun. "Too much phonics." "Not enough whole books." "Time wasted on phonemic awareness." "MTSS doesn't work."

The uncomfortable truth is that structured literacy suffers from the same confusion as "balanced literacy." It remains — in its fine-grained elements — under-defined.[1]

The problem shifts: Enabling pervasive good practices

For decades, reading reform centered on getting evidence-aligned instruction into more than a minority of classrooms.

By 2023, Mark Seidenberg was pressing a different question: now that demand for the science of reading had grown, where was the expertise and infrastructure needed to turn a large, changing research base into dependable practice? Teachers should not have to become cognitive scientists; the science has to be built into the systems around them.

By mid-2026, calls for system integration solutions had appeared from a variety of experts. A panel with Doug Carnine, Sarah Oberle, Mike Schmoker, and Robert Pondiscio offers a video synopsis of these.[11]

In Congress, legislation for an NRP 2.0 has passed out of the Appropriations Committee. If it becomes law, will this solve the problem at hand?

At the scale of millions of teachers and thousands of intermediaries, messages drift as they propagate. Would NRP 2.0 adequately address this?

40% of U.S. fourth-graders read below NAEP Basic in 2024.

Black fourth-graders face massive, systemic hurdles in reading.

From awareness to consistency

2023 · SEIDENBERG[2,3] The science has won attention, but research expertise is thin throughout the education system. Translation and professional infrastructure are missing.

2024–25 · JONES[4,6] A living profession needs "garbage collection"; other professions have solutions that can help cut instructional waste, reduce opportunity costs, help kids read.

2024–26 · CALLS FOR NRP 2.0[5,7,8] In the wake of unsustained and unscaled improvement, calls grow for a new National Reading Panel to synthesize research done since NRP I.

2025–26 · CARNINE/LYON[9,10,11,12,15] frame the problem as professionalization: shared knowledge, common language, aligned preparation, competence-based licensure, accountability.

2026 · TIPTON/PATTON TERRY[13] Beyond the science of reading, we need an "engineering of reading."

THE BEGINNINGS OF PROFESSIONAL INFRASTRUCTURE

The pieces steadily come online

Reading education has begun to acquire much-needed tools and structures.

The Reading League: standards for foundational curricula[14]
Its Curriculum Evaluation Guidelines and Navigation Reports help schools make better choices.

Reading Rockets + Reading Universe[16,17]
Reading Rockets has been refurbished and updated; Reading Universe adds a newer, practical, video-rich body of guidance.

State-built professional learning[18,19]
States are increasingly building grade-, role-, and context-specific training, administrator pathways, and refresher modules.

UFLI brings a free, quality phonics program, with training
Scope and sequence, routines, assessments, videos, fidelity tools, and implementation supports package research into practice.

State-Coordinated Coaching Networks[20]
While 26 states have adopted some mixture of coaching policies, 8 have the type of coordinated field force that brought Mississippi its gains.

ExcelinEd: 18 Early Literacy Fundamental Principles[30]
A state policy blueprint: screening, parent notification, teacher support, intervention, and retention as a last resort.

Learning Science Practices | ResearchEd
Led by our friends in the UK, teachers are tying day to day teaching to learning science well beyond structured literacy.

Fixing the Ed Schools: Teacher-prep standards and audits[21,22]
NCTQ's systematic reviews of teacher-preparation programs has led to state audits and standards for schools of education.

Knowledge Matters: supporting knowledge-building[23]
The Knowledge Matters Campaign focuses on the critical role of knowledge-building in early literacy, and on approaches to do so.

Johns Hopkins ELA Knowledge Map[24]
With the ELA Knowledge Map, reviewers can explore the domains, sequencing, coherence, and gaps in a curriculum's content knowledge.

The EdReports reckoning[25,26]
The exposure of serious weaknesses in the Ed Reports definition of "evidence-based" has led it to release a "2.0 Review Criteria", and to users being far more circumspect about its ratings.

Podcasts and specialist media: solid learning[27]
A strong ecosystem of podcast series and specialist videos now lets those with the inclination learn on their own schedule.

Social-media dispersal and discussion
The Facebook group "Science of Reading—What I Should Have Learned in College" now has 250,000 members. An extremely well-informed X community daily discusses the hard cases of Structured Literacy. Substack gives longer-form ideas immediate feedback.

R&D and systems for older readers, too[28,29]
From more structured fifth-grade word study to systems for severe high school remediation; from individually-authored programs to high end programs like Reading Reimagined, all K12 literacy is getting love.

Building Consistent Nationwide Teaching Capability

No one should doubt the remarkable success represented in the above components. Few, indeed, would have had reason to expect us to have come so far, so fast. That said, practices remain wildly inconsistent (see Appendix).

Still Missing

What's missing is an integrating framework: a unifying structure that ties the components together.

The Gaps — A Case Study

Is Reading Universe a suitable knowledge base? It would appear so: it explains phonemic awareness, decoding, fluency, vocabulary, morphology, and many classroom routines quite well.

Yet critical components are missing: Knowledge-building is not developed as a major segment. Fine-grained issues such as morphophonemic alternation in complex multisyllabic words receive little treatment. Poor practices like teaching "blends" remain.

Further, research traceability is inconsistent and not built into the architecture. Like every resource today, it lacks the maintenance, issue tracking, version control, and feedback processes needed to move the profession forward.

The Road Forward

NRP 2.0 A new National Reading Panel[7,8] is moving forward, yet that 1990s model has aged. In 2026, experts monitor and share new research daily. While a government-sponsored panel may add imprimatur, its static report merely restarts the clock. Other professions offer tools, processes, and cultures to keep guidance current as the evidence changes.

An 'Engineering of Reading' It's a good phrase: an engineering mindset is definitely needed. Today we have more of a writers + publishers mentality driving the system — extremely useful, yet insufficient to the need.
On the other hand, engineering-like documents can only be intermediate steps. Not teacher- or trainer-facing work products. The various engineering professions are all smaller, and generally draw from a more academically elite demographic base than teaching. Where they can suffer more technical jargon, the teaching profession cannot.

An Integrated Field Manual System—Maintained by the Community as a Canonical Source

A Field Manual System

Field manuals are carefully designed publications for people who are not primarily knowledge workers. They are written with the objective of being read by (in Pondiscio's words) mere mortals.

In K12, the focus is thus on the median teacher (with various 'extra duties', a supplemental coaching contract, 3 kids at home, maybe a parent needing care).[31]

Part natural 'field guide' (describing nature), part technical manual (describing a working system and default operating procedures), an SOR field manual would be a science-backed collection of the best all-media resources for learning structured literacy.

A field manual system would offer much more than books published by one or two authors, blog posts, videos, or podcast episodes (not that those don't serve!), and more than a website of independently authored white papers.

It would provide an authoritative snapshot of the profession's knowledge and best practices: evidence-based learning progressions, explicit instructional routines and their limits, assessment and diagnostic tools, decision rules for grouping students and adjusting instruction, guidance for coordinating core instruction and interventions, and procedures for evaluating results and correcting course.

It would (as the military's manuals do) use learning science and instructional principles *in its own design* for teacher/trainer learning.

+

Community-Maintained as Canonical Source

Teaching, in the US, is a far larger and more decentralized profession than most. It has no clear institutional center. Thus, while the US military is a similarly sized profession, and the US Army's doctrine and manual system[32] provides a solid model for a knowledge center, reading education has no TRADOC equivalent to organize and maintain such a canonical source.

There is another similarly sized profession, with similarly decentralized leadership. And they have solved this challenge.

For thirty years, communities of open source software developers have collaboratively built not just products, but documentation and all manner of tools to support the open source processes.

In time, a unique culture also developed to support those people, methods, and tools. Habits of mind actually changed to support the new model.

Reading education can import that culture and process.

- **Processes.** Named maintainers, contribution rules, issue tracking, version control, review cycles, tests, release histories.
- **Culture.** Contribute, don't complain. Many eyes make bugs shallow. Being specifically wrong beats being right and vague.

The resulting knowledge base is more reliable precisely because many capable people can inspect it, challenge it, and improve it — without any one organization holding all the expertise.

Building these is the unfinished work; with them, excellent reading instruction can become commonplace. Teachers and trainers won't have to become reading scientists; their profession will make that knowledge clear, practical, and readily available—in forms that are, if not addictive, at least such that teachers will truly want to use them.

Moving Past Publish & Forget:

Bringing in Teachers & Trainers; Accelerating the Speed of the Loop

Today, guidance is published and left to age. It could be maintained instead.



Let's look at some details of practice:

After K-1 MOY assessments, many teachers are unsure of what to do with the data. → A decision tree is needed. It will look different for different grades and assessment products. → A true expert in the field creates one, and it seems good. It considers changes to Tier I instruction, using additional diagnostics, arranging Tier II intervention, and more.

Then what?
THE TWO MODELS ARE IDENTICAL UP TO THIS POINT

PUBLISH AND FORGET

The decision tree is published as a finished product in a blog post, podcast, book, webinar, etc.
THE DOCUMENT IS THE DELIVERABLE

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Some teachers see it, most don't, few experts review it, and it rarely reaches PD beyond the author's own sessions.

↓

Teachers who do see it find cases it doesn't cover. A few get asked and answered on X or Substack. Maybe on Facebook—where answers run the gamut.
NO ROUTE BACK TO THE SOURCE

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Dead end. Later, another expert writes another decision tree.

OPEN SOURCE ETHOS, TOOLS, PROCESSES

The decision tree is released to a single canonical source, version-numbered and dated.
VERSION CONTROLLED RELEASE

→

Experts and users weigh in, on an open, linked issue tracker. (Feedback isn't lost in evaluation forms or hallway conversations.)
PUBLIC ISSUE TRACKING

↓

A teacher, coach, or specialist proposes a new branch — not just a complaint.
CONTRIBUTION FROM THE FIELD

←

Maintainers accept or decline in public, with reasons on the record; the fix ships as the next version.
NEW VERSION APPROVED

↑

KNOWLEDGE COMPOUNDS

Combinations of guidance can be packaged & published. Local adaptation stays linked to the source instead of drifting away. A good local fix can travel back up. Fidelity and adaptation stop being opposites.

Deeper tracking to the science. Trainers and writers can drill down to linked connections to the research-practice edge.

Nothing is thrown away. Every prior version stays citable. Changes from the 2025 decision tree to the 2028 version are tracked line by line — who changed what, when, and on what evidence.

In the publish-and-forget model, improvement comes via random acts of innovation. In an open source community, improvement is constant and *cumulative:* contributions are reviewed against the evidence, and the best become part of what the profession knows.

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APPENDIX

Where Structured Literacy Practices Can Drift

Even when teachers are committed to structured literacy, their guidance may vary greatly. Some practices below consume time, attention, and working memory without proportional gain—crowding out others that are needed.

PHONEMIC AWARENESS AND PHONICS

Teaching "blends."
A student who knows the sounds for *s*, *t*, *o*, and *p* already has what's needed to read *stop*. While some students need explicit practice blending adjacent consonants like *st-*, *bl-*, and *tr-*, memorizing 20 or more "blends" imposes an undue burden.

Overdoing oral phonemic awareness—for far too long.
Oral PA gains peak after roughly 10 hours of Tier I instruction; PA integrated with letters is more efficacious. So-called "advanced" PA is generally not useful.

Sound walls: teaching articulatory terminology beyond its instructional payoff.
Mouth pictures and articulation cues can help students tell similar sounds apart. Drilling terms like *bilabial*, *fricative*, or *voiceless alveolar stop* rarely improves reading.

Teaching syllable division as a rule system rather than a flexible aid.
Students may spend excess time marking VC/CV, V/CV, or VC/V divisions as if they determine pronunciation. Stress, morphology, and vowel flexibility often matter more.

Treating six syllable types as a decoding algorithm.
Open, closed, vowel-consonant-e, r-controlled, vowel-team, and consonant-le categories can be useful descriptions. Teachers often overdo them.

Teaching phonics patterns without enough connected-text reading.
Patterns stick best, and words get orthographically mapped, when students meet them repeatedly in connected text.

Inadequate or misaligned spelling instruction.
Encoding—spelling the patterns students are learning—is one of the strongest supports for reading.

MORPHOLOGY AND ORTHOGRAPHY

Ignoring morphology until students are considered "advanced."
A student may be taught to attack *unhelpfulness*, *disagreement*, or *reconstruction* primarily through syllabification even though meaningful units provide a clearer structure. Delaying morphology can make complex words unnecessarily difficult.

Introducing morphology as lists of prefixes and suffixes.
The opposite error also occurs. Students memorize *pre-*, *re-*, *un-*, *-ful*, *-tion* without learning how morphological structure explains meaning, spelling, word families, or pronunciation changes.

Failing to teach morphophonemic change.
In pairs such as *define/definition*, *sign/signature*, *serene/serenity*, and *electric/electricity*, pronunciation shifts while spelling preserves the shared meaning. This helps explain "silent" letters and unclear vowels (*compete → competition*), shows why English spelling cannot be understood solely as sound-to-letter matching, and keeps regular morphological spellings from being taught as "exceptions."

Using "schwa" as an explanation rather than teaching what causes it.
Students may be told that an unstressed vowel "says /uh/" without learning the role of stress, morphology, and word structure. The label names the phenomenon but does not necessarily help them read or spell the word.

Teaching too many heart words or "irregular" word parts.
Words like *have* (English words don't end in *v*) and *two* (*twin*, *twelve*) are often taught as heart words, though their spellings are explainable. When a word breaks an introductory rule, the cause is often stress, morphology, history, or a more advanced correspondence. Calling each case an exception adds memorization and makes English look less systematic than it is.

Overteaching linguistics or etymology.
Explanation can also go too far: rabbit holes into word origins and linguistic processes that don't pay off. An explanation earns its place when it helps students read or spell many words, not just one.

TEXT SELECTION

Equating decodability with instructional suitability.
A text can conform to taught correspondences and still impose awkward syntax, weak vocabulary, low coherence, or excess cognitive load. Conversely, excluding every word containing an untaught pattern can impoverish language and knowledge.

Using controlled text for too long.
Decodable text can be valuable while students consolidate the alphabetic system. If tightly controlled text persists after students can handle broader print, it may constrain vocabulary, syntax, fluency, and knowledge-building.

Moving students out of controlled text too early.
The reverse error leaves students guessing from context because the text contains too many patterns they cannot yet decode reliably.

FLUENCY

Treating oral reading fluency as a single problem.
A low words-correct-per-minute score, or accurate but slow reading, can reflect weak decoding, automaticity, multisyllabic word skill, phrasing, vocabulary, content knowledge, or print exposure. Locate the source before choosing a response.

Accurate decoding ≠ automatic word recognition.
A student who decodes a word correctly may need many more encounters before recognizing it instantly. Check speed as well as accuracy.

Ignoring prosody when assessing fluency.
Appropriate phrasing, pausing, stress, and intonation provide information that accuracy and rate alone miss. Prosody relates meaningfully to comprehension.

Using repeated reading when the bottleneck is decoding.
A student can become faster at a practiced passage without acquiring the word-reading knowledge needed to transfer that improvement to a new passage.

Using round-robin or popcorn reading for oral reading practice.
Each student reads only a few sentences while classmates wait, read ahead, or tune out, and weak readers are exposed rather than supported. Partner, choral, and echo reading give every student far more practice.

VOCABULARY, KNOWLEDGE, AND COMPREHENSION

Underteaching oral language.
Many students lag in spoken language. Listening comprehension should be explicitly built from K up, via read-alouds, discussion, prompted storytelling, and extended talk.

Teaching vocabulary as isolated definitions.
Students may copy definitions, match words to meanings, or memorize weekly lists without learning semantic relationships, morphology, multiple meanings, collocations, or the knowledge domain in which the words are useful.

Underteaching academic vocabulary.
Words such as *analyze*, *contrast*, *infer*, *significant*, and *establish* occur across subjects and carry much of the meaning in textbooks, directions, assignments, and tests.

Trying to activate knowledge instead of building it.
Students cannot activate knowledge they do not have.

Teaching comprehension through excessive strategy rehearsal.
Predicting, questioning, summarizing, and identifying main ideas can help students learn how comprehension works. Once students understand a strategy, more practice yields little. The time is better spent building vocabulary and knowledge.

Teaching "main idea" as though it were independent of knowledge.
Students may be asked to infer main ideas from passages about topics they barely understand, making a knowledge deficit look like a strategy deficit.

Underteaching sentence structure.
Students may know every word in a sentence and still misunderstand who did what to whom, which idea modifies which, or how one clause relates to another. Passive voice, relative clauses, embedded clauses, and connectives such as *although*, *unless*, and *despite* can trip up many learners.

Underteaching text structure.
A text's organization (comparison, cause-effect, problem-solution) shows how its ideas connect. Students who learn to spot it from signal words (*because*, *in contrast*) can better grasp the gist and remember what they read.

Overteaching grammatical labels.
Students can spend substantial time naming predicates, subordinating conjunctions, or clause types without getting better at understanding or writing hard sentences.

PROGRAMS, ASSESSMENT, AND INTERVENTION

Assuming a program scope and sequence = a knowledge model.
A commercial program necessarily makes sequencing choices. Those choices may be sensible, but teachers can begin treating one program's terminology, order, and routines as though they represented settled findings about reading development.

Treating fidelity as immunity from error.
Implementation fidelity is useful only if the underlying practice is well matched to the student and the evidence. Perfectly implementing the wrong response simply produces consistent inefficiency.

Assigning intervention when Tier I instruction needs improvement.
If more than ~20% of students are struggling, the grade-level team should review core instruction wrt pacing, content, sequencing, instructional time, and delivery.

Three tiers does not mean three programs.
Unrelated programs at each tier give struggling students conflicting routines and terminology. Intervention works best when it reinforces core instruction.

Responding to every error with immediate correction.
Constant interruption undermines fluency and comprehension. Teachers need guidance on which errors matter and when to step in.

Treating a screening score as a diagnosis.
Two students with the same screening score may struggle for very different reasons: weak phoneme-grapheme knowledge, slow automatic recognition, language disorder, limited vocabulary, etc. Good diagnostic assessments should precede intervention; core instruction may also need adjustment.

Undiagnosed DLD.
More than 7% of kids have Developmental Language Disorder. The earlier it's diagnosed, the better. Yet many schools don't screen early enough, treating language difficulties as generic reading problems.

Continuing "evidence-based" instruction without checking response.
Population-level evidence does not guarantee that a particular routine remains useful for a particular student. Instruction should change when progress data show that the presumed mechanism is not producing the expected result.

> This appendix contains significant AI text. It is the authors' hope that you, the experts, will embrace this as a community, and build it to better tell the story.